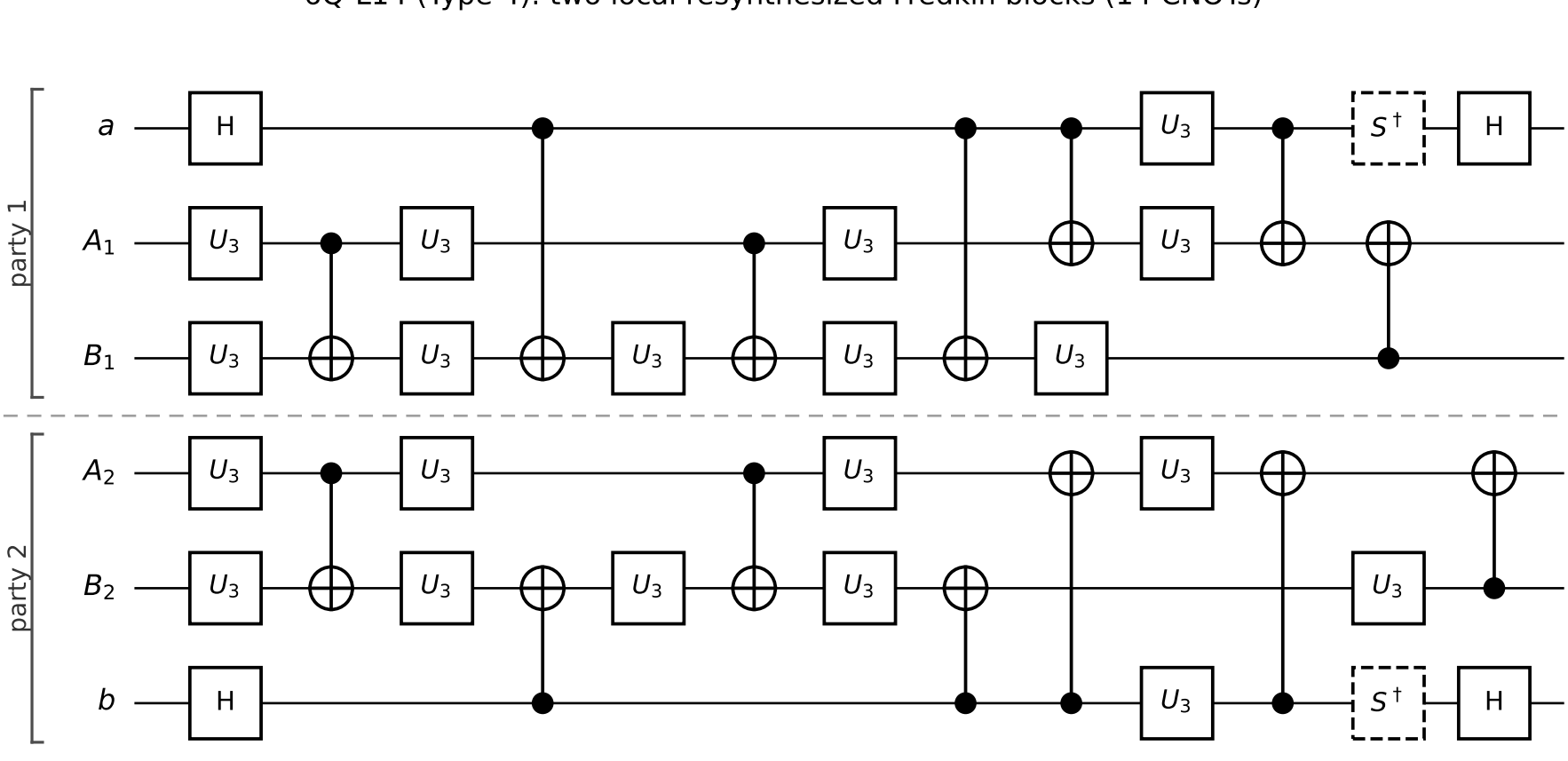
6Q-L14 (Type 4): two local resynthesized Fredkin blocks (14 CNOTs)
party 1
party 2
a
A1
B1
A2
B2
b
H
U3
S†
Each party runs the seven-CNOT block $G_7 = e^{-i\pi/8}F$ extracted from the Step-4 gate list (Qiskit, basis $\{u, \mathrm{cx}\}$);
no two-qubit gate crosses the party cut. A two-qubit replacement depolarizing channel D_q is applied after every CNOT.
Dashed $S^\dagger$: present only in the YY setting. The estimator $\Delta(O) = \langle X_a X_b \otimes O\rangle - \langle Y_a Y_b \otimes O\rangle$ requires both settings.